A Systematic Evaluation of a Cyclical Bull Call Spread Strategy

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1 Abstract

Investing is fundamentally based on the appropriate management of risk versus reward. Options investing enables the construction of non-linear payoff structures which demand more deliberate risk management than traditional equity exposure. This study constructs and systematically backtests a defined bullish call options spread strategy across a filtered universe of equities within a synthetic Black–Scholes options pricing framework. The strategy is evaluated over a multi-year horizon with emphasis on reward-to-risk constraints, capital efficiency, and pricing ranges. Although the options spread constructor is based on key optimized variables, the results demonstrate that structural strategy design alone does not ensure sustainable growth. A broader insight can be understood as a consequence of this study: in options-based strategies, effective risk management extends beyond the design of the trade itself and is critically shaped by how risk is managed throughout the life of the position.

2 Introduction & Motivation

When making investment decisions of any kind, proper consideration of risk and reward is essential. From those who purchased farm land in the late 1800s to those expanding their digital wallet in the current crypto climate, proper risk management revolving around investment decisions have never wavered. At its core, any competent investor should always wrestle with the following: *What return can be expected and how likely is success? If the results are unfavorable, how much do I risk losing?*

In public equity markets, these questions manifest in a relatively transparent manner. The focus is linear – if the underlying asset increases in price, the value of one's position increases proportionally; conversely, if the stock declines, one's position decreases in value with the maximum loss being realized only if the stock goes to zero. Risk and reward scale symmetrically and an investor's primary concern is whether the expected appreciation of an asset justifies a potential downside risk relative to the original purchase price.

However, as financial markets have evolved, the development of derivative instruments has transformed how these fundamental questions are addressed. A 'derivative' - is a secondary bet the value of which is derived from the price of the underlying security. Among these instruments, options in particular, have emerged as a powerful tool offering significant benefits but requiring investors to rethink risk management and traditional approaches to public equity investing.

An option is, at its most basic level, an agreement between two parties for the sale or purchase of some asset, at an agreed upon price, over a limited period of time. Unlike traditional stock transactions, option trades exhibit non-linear risk and reward patterns shaped primarily by the strike price, time to expiration, premium paid, and the volatility of the underlying asset. Options take two primary forms: calls and puts. A call option grants the holder the right to buy the asset at a certain price, while a put option grants the right to sell. Although each instrument may be useful by itself, given a certain market view, combining the buying and selling of options, each transaction known as a leg often provides traders with greater flexibility and better control over risk exposure. It is with this framework in mind that this paper seeks to construct and then backtest a specific options spread strategy over a five-year horizon.

3 Strategy Definition

3.1 Instrument

A vertical bull call spread involves the simultaneous buying and selling of call options of the same asset at distinct strike prices with the same expiration date. The bought ("long") call option is entered into at a lower strike price, while the sold ("short") call option is at a higher strike price, with the difference between the strike prices being the width of the spread. Traders who feel the underlying asset will appreciate in price but wish to wish to decrease the cost of the trade and thereby limit the downside risk due to a potential incorrect prediction typically enter this trade. The defined downside is limited to the net debit - or cost of the spread, while its upside payoff is essentially the spread width minus debit paid (Max Profit = W - Net Debit Paid). Given the specific strategy being tested employs a bull call spread across all selected stocks it requires a moderately bullish directional view. Reward-to-risk ratio (RRR) is a foundational parameter which plays an integral role throughout the trade entry logic design. This trading metric compares potential profit of a trade to its potential risk. In the case of our strategy the RRR is essentially the Max Profit divided by Net Debit Paid (RRR = MP/D).

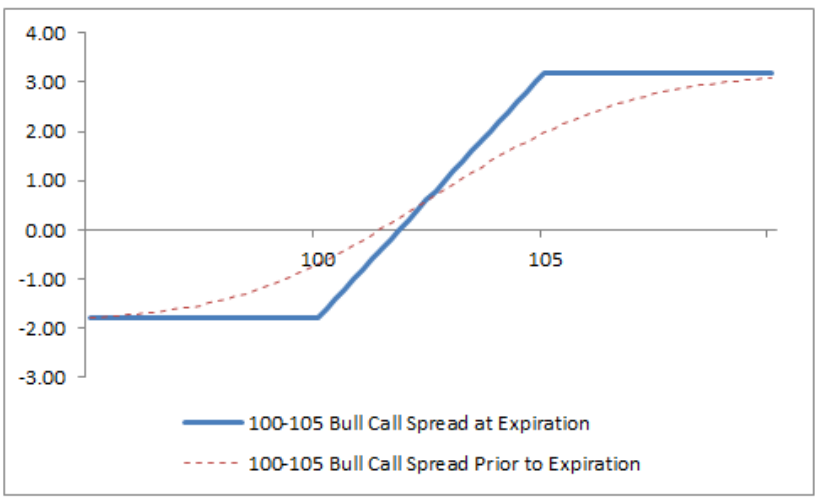

Figure 1: Bull Call Spread Payoff Diagram

3.2 Universe Selection

The strategy is implemented on a select number of stocks, which have been filtered through two phases: Phase 1 (Stock Filter) and Phase 2 (Option Filter).

Phase 1

Using FinViz, the following quantitative constraints were applied to an initial universe of approximately 7000–8000 publicly traded equities:

1. $P > SMA_{200}$
2. $SMA_{50} > SMA_{200}$
3. $ADV_{30} \geq 1{,}000{,}000$
4. $P > \$10$
5. $M > \$2\,B$

6. $\text{DTE}_{earn} \geq 14$,

where P denotes the stock price, SMA_{200} and SMA_{50} denote the 200-day and 50-day simple moving averages, respectively, M denotes a company's market capitalization and DTE_{earn} denotes the number of calendar days until the next earnings announcement.

Phase 1 of the Universe Selection process aims to narrow FinViz's equity pool to stocks with a relatively strong positive trend, as indicated by the technical indicators. While ensuring compatible liquidity, price (i.e. no penny stocks), and an earnings filter to eliminate stocks which are less than or equal to 14 days away from company earnings (to avoid high volatility climates prior to the Earnings Date). Based on this filtration process a total of 585 equities remained.

Phase 2

1. $OI_{strike} \geq 500$
2. $OV_{strike} \geq 50$
3. $5 \leq OI/OV \leq 50$
4. $BAS \leq 0.10$,

where OI_{strike} denotes the open interest at a given option strike, OV_{strike} denotes the traded option volume at that strike, OI/OV represents the open-interest-to-volume liquidity ratio, and BAS denotes the option bid–ask spread (in dollars).

Phase 2 seeks to further narrrow this pool by restricting the strategy universe to stocks whose call options near 60 days-to-expiration (DTE) not only exist, but more importantly meet minimum liquidity and cost thresholds. This phase ensures the equity pool is optionable and has sufficient liquidity to support a practical implementation of the strategy. This phase reduced the original pool down to 64 equities.

3.3 Spread Construction Rules

The spread aims to provide the trader with capital-efficient bullish exposure while maintaining a realisitic probability of positive payoff. In options trading delta (δ) is a key risk metric which estimates how much an option's price will change for every $1 dollar shift in the underlying asset.

This strategy aims to construct bull-call spreads with the selected universe of stocks which have the long call slightly to moderately in-the-money (ITM, $0.55 \leq \delta \leq 0.70$) and the short call out-of-the-money (OTM, $0.20 \leq \delta \leq 0.40$). Selecting the long call in the desired delta range provides a moderately favourable, risk-neutral probability of expiring ITM, assuming reasonable volatility. The strategy targets approximately 60 DTE as the optimal time horizon for each spread, while allowing a bounded range (25-80 DTE) thereby accomodating option expiry availability and other constraints.

A spread trade consists of two components: spread width and net debit paid, with both directly tying to the trade's RRR. Originally the perspective of a desired RRR of 1.75 resulted in a constraint: $D \leq 0.36 * W$, where D represents the net debit and W the spread width. This condition ensured attractive risk–reward profiles; however, it proved overly restrictive in practice. Due to live option pricing and prevailing market conditions, virtually no actively traded spreads within the selected equity universe satisfied this requirement. Consequently, the framework was modified to incorporate a DTE-dependent maximum debit as a fraction of spread width. This adjustment materially increased the number of viable spreads while preserving disciplined risk–reward characteristics.

Additional filters are added to ensure that the constructed spreads reflect realistic probability and execution constraints. To start, a breakeven constraint is enforced such that the breakeven price at expiration stays within an appropriate range above the current stock price. This requirement prevents spread possibilities which require a big upward move- in this case 8%- in the underlying asset to begin profiting. Next, a short strike distance filter is added. Spreads where the short strike is greater than

10% above the current price are rejected, thereby increasing the probability of reaching maximum profit. If the short strike is too far away, one may still profit, but there is a lower likliehood in realizing the full payoff. Third, the spread constructor avoids extreme pricing regimes, where the debit fraction (D/W) is either too high or too low. This keeps spreads from either being too cheap where there's often low probability of profit, or cases where one is paying most of the width and probability to profit is high, but the payoff is not as attractive.

Finally a minimum absolute width of $3 is implemented, therefore avoiding tiny spreads which are fragile in practice. In general, these filters serve as practical bounds for probability and trade quality, ensuring that optimal ITM trades are constructed with realistic breakeven levels and attainable upside targets under conservative execution assumptions. The program, for a given underlying security and expiration date, ensures that all candidate spreads satisfy these constraints and then evaluates the filtered spreads based on RRR. Among these feasible candidates, the spread with the highest RRR is selected as the optimal construction. Although no explicit delta constraint was ever enforced, approximately 70% of spreads fell within the desired delta range. The remainder of the spreads fell relatively close to, or just outside the range. Emergence of these delta characteristics is due to structural constraints the spreads were built upon. The fact that a specific delta constraint was not directly enforced, and yet the desired range was met, is a testament to the strong quantitative design of the strategy.

4 Methodology: Synthetic Backtest

4.1 Objective & Scope

The objective of this backtest procedure involves testing the implementation of this strategy over a 5-year window, from the beginning of 2021 until the end of 2025. A synthetic backtest involves the creation of past options prices based on real historical stock data. It enables the evolution of a strategy in a controlled environment allowing a consistent comparison across time and assets. Also, at times historical option chains are incomplete and illiquid with past data being difficult to access. Synthetic backtesting therefore solves this problem by creating synthetic historical prices while simultaneously isolating the structural performance of the strategy from market frictions innate within realized option data. As mentioned this backtest is conducted within a 'controlled environment' that relies on key assumptions, much of which heavily simplifies real-world market conditions. The key assumptions are:

1. Implied volatility (IV) is assumed to be constant across strikes and maturities at the time of entry.
2. No change in volatility occurs since the position is entered into up until the option expires.
3. All positions are held until expiration.

4.2 High-Level Pipeline

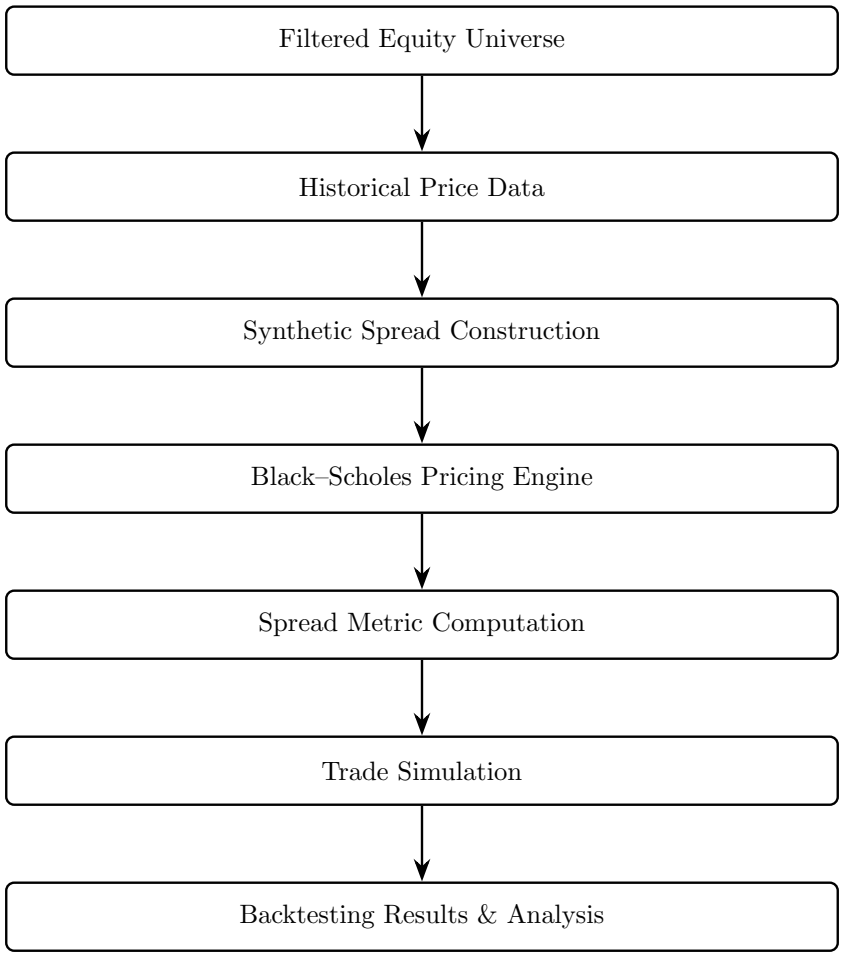

Figure 2: High-level backtesting pipeline.

4.3 Synthetic Option Pricing Framework

Once an optimal spread is constructed the short strike, K_{short} and long strike, K_{long} values are confirmed. The next step involves creating synthetic historical option prices based on the inputted strike prices.

Part 1: Theoretical Option Valuation

Given a set of strikes and various market inputs, the first part simply computes a fair, theoretical price of a European call option (exercised only at expiration) by using the Black-Scholes Model (BS). This model essentially takes into account the volatility of a stock and how much time remains until expiration, and attaches a fair price today for the value of its call option (i.e. the right to buy that stock at a fixed price in the future). The spot price, S is the current underlying price of the stock. The strike K, is the fixed, pre-determined price at which the trader can exercise said option (this is the outputted figure of the spread constructor).

Time to expiration, T, refers to how many days are left until expiration, and risk-free rate, r, is the minimum return a trader expects on an investment with zero risk, acting as a benchmark for all other investments. For the purposes of this project a constant 3% annualized rate is assumed. Finally volatility, σ, refers to the market's expectation of how much a given asset will fluctuate in price over a period

of time. In this study, volatility is estimated using 20-day realized volatility computed directly from historical price data. This measure reflects the stock's recent price behaviour over approximately one trading month. Given inputs of the spot price, strike price, time to expiration, risk-free rate and volatility the Black-Scholes formula outputs the mid-price of one fair theoretical call price. In total two call prices are outputted, one per strike, for a given stock.

Under the Black–Scholes framework, the theoretical price of a European call option is given by:

$$C(S, K, T, r, \sigma) = S\,N(d_1) - Ke^{-rT}N(d_2), \tag{1}$$

where,

$$d_1 = \frac{\ln(S/K) + \left(r + 12\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}. \tag{2}$$

This function outputs the historical options price for each strike, therefore producing two mid-prices, one for the bought call and the other for the sold call.

Part 2: Bid-Ask Pricing Implementation

Although the Black-Scholes model produces a theoretical mid-price, option transactions in the real marketplace occur at bid and ask prices. Bid refers to the highest price a buyer is willing to pay, while the ask is the lowest price a seller is willing to accept. For each call option, the theoretical price is adjusted by a fixed proportional spread factor, thereby generating synthetic bid and ask prices. A long call is assumed to be purchased at the ask price, while the short call is sold at the bid price. The net debit paid by the trader is simply the difference between the long-leg ask price and the short-leg bid price. The figure for debit paid represents the full amount the trader risks losing once entering into this trade.

4.4 Synthetic Backtesting Procedure

The backtest procedure simulates an execution of trades in a routinely fashion for a given security over a certain period of time. In this study, filtered stocks were traded through a five-year lifecycle (2021-01-01 to 2025-12-31), with cyclical 60 DTE bull-call spread trades placed consistent with the defined constraints and rules within the spread constructor file. The backtesting process begins on the first trading day of 2021 and ends on the last trading day of 2025 (5-year period). However, the first trade is initiated only after sufficient historical data is available to estimate realized volatility (20 trading days), resulting in an initial entry in early February 2021.

Trades are held until expiration, which is essentially the full time the option contract is viable for. This condition, an enforced 'hold-to-expiration' rule, increases risk substantially as the final days preceding expiration are where huge swings in profit and loss occur. For example, a trader who feels they are in a losing position can contain substantial losses and exit their position taking a minor hit, instead of losing the full debit paid. For the purpose of this backtesting process the original rule will stand. Once the ~ 60 DTE are up, for a given stock the next trade is placed immediately on the subsequent trading day.

For instance, a bull call spread trade on AAPL was entered into on the date: 2021-02-02 (February 2nd, 2021) and the trade was exited on 2021-04-01 (April 1st, 2021). The next trade was placed on Monday, April 5th, 2021— as April 2nd was 'Good Friday' and U.S. markets were closed. This results in a non-overlapping trade design which means only one trade is active at a time per ticker. A key point to recall: once a position is entered, implied volatility is assumed to remain constant over the life of the trade.

As positions are held to expiration, no repricing due to changes in volatility is taken into account. This is an assumption which simplifies the backtesting procedure, making it distinct from real-world scenarios. Strategy performance is evaluated using both return on capital at risk (debit) and return relative to maximum spread width. This entire procedure produces a time series of distinct spread outcomes,

enabling a thorough evaluation of win rates, return distribution, cumulative performance and overall success of the strategy. All performance metrics and empirical results derived from this framework are presented and analyzed in Chapter 5.

5 Results & Performance Evaluation

This chapter seeks to answer: *What results did our backtest produce?* Prior to delving into successfully backtested trades, many stocks, although present in the filtered list, were never backtested at all. Certain securities were excluded due to insufficient continuous price history to compute the 20-day realized volatility required by the model. This may occur for recently IPO'd stocks, securities with incomplete data, or those experiencing structural breaks in their price series. Without a stable volatility estimate, synthetic option prices cannot be generated reliably, and the backtest is therefore not executed for those securities. In addition, even when sufficient price history is available, the synthetic Black–Scholes pricing framework may produce option chains that do not permit the construction of economically viable spreads. Specifically, certain securities fail to generate bull call spreads satisfying the model's constraints, including minimum reward-to-risk ratio (RRR), acceptable debit-to-width fraction, and structural width requirements. In such cases, the strategy refrains from entering a position rather than forcing exposure under unfavorable payoff conditions.

Distribution of Returns

The segment of this chapter seeks to analyze the per-trade results of this backtest over a 5-year window. The analysis seeks to answer the pertinent question of: *What does one bull call spread look like, probabilistically?* Hence, the key metric of focus is the return on debit for each trade, which is:

$$R_{debit} = \frac{Intrinsic_T - D}{D} \tag{3}$$

where R_{debit} is the return on capital invested, $Intrinsic_T$ is the intrinsic value of the spread at expiration, and D is the initial debit paid. For example, if both calls expire worthless, the entire debit is lost, meaning the spread's intrinsic value at expiration is 0 thus producing a $R_{debit} = -1$, or -100%. Figure 3 illustrates the distribution of per-trade returns over the five-year backtest.

Based on the histogram, specific key points can be extracted. To start with, the backtest generated 915 trades between 2021 and 2025. Of the 915 trades 41.5% resulted in a full loss of invested capital. In other words, 380 trades had the spread value expire worthless, thus producing a R_{debit} of -100%.

The outcome where the full debit paid is lost is inherent to the payoff structure of a bull call spread. In the cases where the stock finishes below the long strike at expiration, the initial debit is forfeited as both legs expire worthless. It is important to note when backtesting the strategy, we have enforced a rule to hold all trades until expiration, therefore excluding any tactics to mitigate losses. In a real-world scenario, a trader may close their spread position once a certain percentage of the debit paid is lost, or a specific number of days before expiration, in order to prevent a full loss. Given risk management tactics are excluded from the backtest, the 41.5% full-loss frequency should be interpreted as a conservative estimate of downside exposure under a strict hold-to-expiration framework.

The median return on debit paid across all trades is -24.14%, the mean return is 12.09%, and the overall win rate stands at 47.43%. The median return provides insight into what the "typical" trade looks like, while mean return defines the strategy's expected value over the full sample set. A negative median return indicates more than half of all trades were unprofitable. At least 50% of trades produced a loss of -24.14% or worse, while the remaining 50% resulted in a return better than -24.14%. With that said, given the expected payoff remains positive, this positive expectancy is driven by the magnitude of the wins rather than the frequency. In other words, even though majority of trades lose money, the profitable trades produce gains of sufficient size to offset the majority losses. Specifically as shown on the Distribution Chart (Figure 3), a significant number of trades has a RRR greater than or equal to 100%.

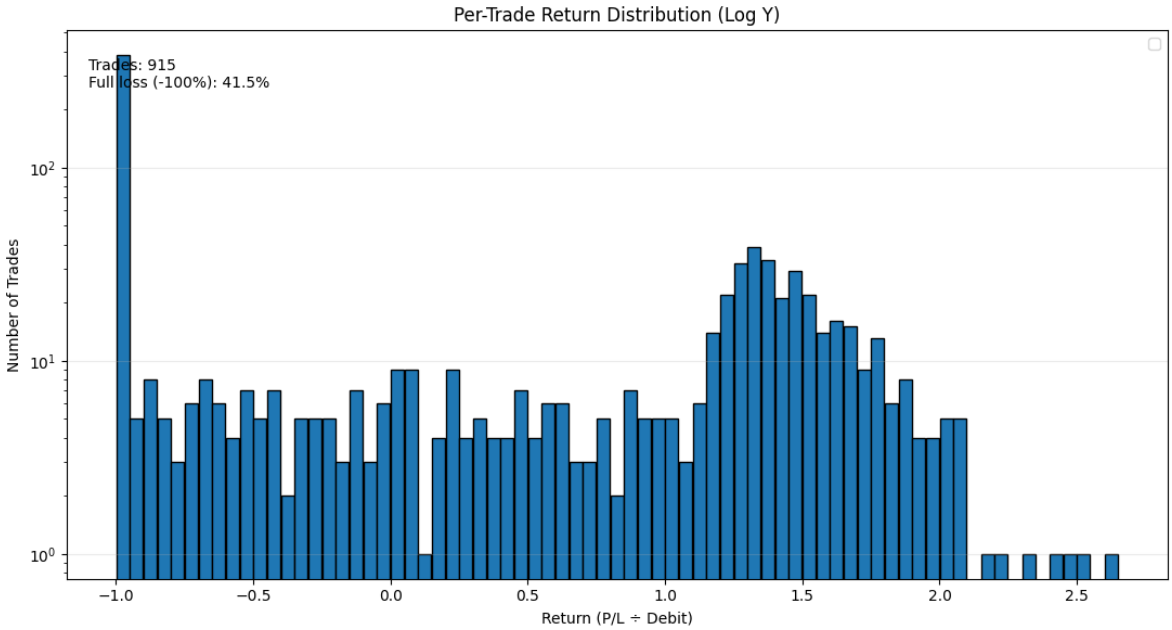

Figure 3: Distribution of Return on Debit for Bull Call Spreads (2021–2025)

6 Conclusion

This paper set out to construct and evaluate a well-defined, risk-conscious, directionally bullish options strategy within a synthetic pricing framework. Grounded in the fundamental principle that investing revolves around adequately balancing risk and reward, this analysis sought to determine whether this trading strategy could generate profitable performance over a 5-year period. Given that the backtest showed a Mean RRR over all trades of +12.09%, it can be concluded that a Bull Call vertical debit spread strategy can generate an attractive RRR-Reward Risk Return!

With that said, the results of the backtest do reveal a central insight: effective risk management extends beyond the inherent construction of the spreads themselves. Although the bull call spreads were designed with reward-to-risk ratio, capital intensity, and price constraints at the forefront of considerations, the findings suggest that disciplined implementation and active efforts by the trader to mitigate risk is just as important as the construction of the trade itself.

Notably, the backtest imposed a strict hold-to-expiration rule, thus isolating the intrinsic payoff characteristics of the strategy. In practice, discretionary risk overlays—such as predefined capital loss thresholds, early liquidation when adverse price movement accelerates, or exiting prior to expiration—may materially influence the outcomes achieved. Ultimately, this study reinforces a broader lesson: in options-based investing, the architecture of risk and reward is only the beginning. Sustainable, profitable performance depends not only on how trades are constructed, but on how risk is continuously managed once exposure is established.